Cost Function

What is Cost Funnction?

- A cost function is a way of measuring the success of an ML project. It is a numeric value that indicates the success or failure of a particular model without needing to understand the inner workings of a model. This is useful in many situations, such as when an engineer wants to know if the cost of building a model is worth it. Also, they can use a cost function if they want to compare different models or versions of the same model. In any case, where it’s essential to know if a project was successful or not, cost functions can be beneficial.

Why is cost function needed?

- Basically, our machine learning model predicts the new value. We aim to have the predicted value as close as possible to the predicted value. If the model predicts the value close to the actual value, then we will say it’s a good model. So here is where cost function plays an important role. This cost function is needed to calculate the difference between actual and predicted values. So here it is nothing, just the difference between the actual values-predicted values.

Cost Function(contd.)

- Cost function=(actual values-predicted values).
- Which is also represented as
- Cost function(d)=Y-Y’
- Where Y=Actual value
- Y’=Predicted value

Cost Function(contd.)

- Similarly, is we have to find this cost function for all data points, then we can write like this:
- Cost function0(d0) =Y0-Y0’
- Cost function1(d1) =Y1-Y1’
- Cost function 2(d2) =Y2-Y2’
- Cost function3(d3) =Y3-Y3’

Cost Function(contd.)

- Cost function n(dn) =Yn-Yn'

So we can use formula to calculate cost function for all points

$$minimize_{\theta_0,\theta_1} \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

- Where
  - $h_\theta(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$
  - $(x^{(i)}, y^{(i)})$ is the i^{th} training data
  - m is the number of training example
  - $\frac{1}{2}$ is a constant that helps cancel 2 in derivative of the function when doing calculations for gradient descent

Cost Function

So, **cost function** is defined as follows,

$$J(\theta_0, \theta_1) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y^{(i)}} - y^{(i)}\right)^2 = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2$$

The line fitting the data points as shown in fig is the result of the parameter Θ0 and θ1.Therefore, the goal of any learning algorithm is to find, or choose, the best parameters that fit the dataset.

Calculate the error between the predicted and expected values according to the formula.

NOTE: The point to note here is that the lesser the value of the cost function, the more accurate our model is, and the more the value of the cost function, the less accurate our model will be.

Types of a cost function in linear regression

Regression cost function:

- Mean error
- Mean squared error
- Mean absolute error
- Root mean squared error (RMSE)

Regression models are used to predict continuous variables, Such as. Employee salaries, car costs, availability of loans, etc. “regression cost function” is the cost function used in regression problems. Depending on the distance-based error, it is determined as follows:

error = y-y`

Y – actual input and Y` – predicted output.

For obvious reasons, this cost function is also known as the squared error function. It is the most commonly used cost function in linear Regression because it is simple and works well.

Mean error

- This cost function computes the error for all training data and derives the average of all these errors. Computing the mean of the errors is the most straightforward and most intuitive method possible.

- These errors can be negative or positive. Therefore, they can cancel each other out during the summation, and the average error of the model will be zero.

- Therefore, it is not the recommended cost function but is the basis for other regression model cost functions.

Mean squared error

- Mean squared error is one of the most commonly used and earliest described regression measures. MSE represents the mean square of the difference between the prediction and the expected result. In other words, MSE is a variation of MAE that squares the difference instead of taking the absolute value of the difference. There is no possibility of negative errors.

$$\text{MAE} = \frac{1}{N}\sum_{j=1}^{N}\left(Y_j - Y_j'\right)^2$$

Where:

Yj: actual value

Y^ j: predicted value from the regression model

N: number of datum

Mean absolute error

Mean absolute error is a regression metric that measures the average error size for a group of predictions without regard to direction. In other words, it is the average absolute difference between the prediction and the expected result, with all individual variances of equal importance. The Mean Absolute Error (or MAE) tells the average of the absolute differences between predicted and actual values. By calculating MAE, we can understand how wrong the model did the predictions.

$$\text{MSE} = \frac{1}{N} \sum_{j=1}^{N} |(Y_j - Y_j')|$$

Mean absolute error

- Where:

Y_j: actual value

$\hat{Y}_j$: predicted value from the regression model

N: number of datum

Root Mean Square Error

It is the square root of the mean of the square of all of the errors. Root Mean Square Error (RMSE) measures the error between two data sets. In other words, it compares an observed or known value and a predicted value.

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(S_i - O_i\right)^2}$$

Where

Oi=observations

Si= predicted values of a variable

n =number of observations

Conclusion:

- The cost function serves as a monitoring tool for various algorithms and models as it shows the difference between predicted and actual results and helps improve the model.